Projected Fisher SBM

1 Models

Consider a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with nodes $\mathcal{V}$ and edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. The adjacency matrix $\mathbf{A}$ is defined such that $\mathbf{A}_{uv} = 1$ if $(u, v) \in \mathcal{E}$ and 0 otherwise. We assume that $\mathcal{G}$ is generated by a Stochastic Block Model (SBM), so that each node $u \in \mathcal{V}$ belongs to one of k communities.

Community membership is defined by the partition matrix $\mathbf{Z}$ such that $\mathbf{Z}_{ui} = 1$ if node u is a member of community i and 0 otherwise. The structure matrix $\mathbf{\Theta}$ is defined such that $\mathbf{\Theta}_{ij}$ is the expected number of edges (or edge weight) from nodes in community i to nodes and community j. That is, each element of the adjacency matrix is generated iid with conditional expectation $\mathrm{E}(\mathbf{A}_{uv}|\mathbf{Z}) = \mathbf{Z}'_u \mathbf{\Theta} \mathbf{Z}_v$.

1.1 Bernoulli

In the simple definition of $\mathcal{G}$ (see above), the adjacency matrix $\mathbf{A}$ only takes elements in $(0, 1)$. Therefore, this type of graph is well-described by the assumption

$$\mathbf{A}_{uv}|\mathbf{Z} \sim \mathrm{Bernoulli}(\mathbf{Z}'_u \mathbf{\Theta} \mathbf{Z}_v)\,.$$

The log-likelihood of this model is

$$\begin{aligned} \ell(\mathbf{Z}, \mathbf{\Theta}|\mathbf{A}) &= \sum_{u,v} \big[\mathbf{A}_{uv} \ln\!\big(\mathbf{Z}'_u \mathbf{\Theta} \mathbf{Z}_v\big) + (1{-}\mathbf{A}_{uv}) \ln\!\big(1{-}\mathbf{Z}'_u \mathbf{\Theta} \mathbf{Z}_v\big)\big] && (1) \\ &= \sum_{u,v} [\mathbf{A}_{uv} \ln(\mathbf{P}_{uv}) + (1{-}\mathbf{A}_{uv}) \ln(1{-}\mathbf{P}_{uv})] && (2) \end{aligned}$$

where $\mathbf{P}_{uv} = \mathbf{Z}'_u \mathbf{\Theta} \mathbf{Z}_v$. The gradient of this log-likelihood is

$$\begin{aligned} \frac{\partial \ell}{\partial \mathbf{Z}_u} &= \sum_v \left[\frac{\mathbf{A}_{uv} - \mathbf{P}_{uv}}{\mathbf{P}_{uv}(1{-}\mathbf{P}_{uv})}\right] \mathbf{\Theta} \mathbf{Z}_v && (3) \\ &= \mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} (\mathbf{A}_u - \mathbf{P}_u) && (4) \end{aligned}$$

where

$$\mathbf{W}^{[u]} = \mathrm{diag}\left(\dots \frac{1}{\mathbf{P}_{uv}(1{-}\mathbf{P}_{uv})} \dots\right)\,. \tag{5}$$

Note that the weight matrix $\mathbf{W}^{[u]}$ is defined over nodes $v \in \mathcal{V}$ with fixed u. The hessian of this log-likelihood is

$$\frac{\partial \ell^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_u'} = -\sum_v \left[\frac{\mathbf{A}_{uv} - \mathbf{P}_{uv}}{\mathbf{P}_{uv}(1-\mathbf{P}_{uv})} \right]^2 \mathbf{\Theta} \mathbf{Z}_v \mathbf{Z}_v' \mathbf{\Theta}' \tag{6}$$

$$= -\mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} \mathbf{S}^{[u]} \mathbf{W}^{[u]} \mathbf{Z} \mathbf{\Theta} \tag{7}$$

where

$$\mathbf{S}^{[u]} = \mathrm{diag}\big(\ldots (\mathbf{A}_{uv} - \mathbf{P}_{uv})^2 \ldots \big) \,. \tag{8}$$

Note that the squared-error matrix $\mathbf{S}^{[u]}$ is defined over nodes $v \in \mathcal{V}$ with fixed u. The expectation of the hessian is

$$\mathrm{E}\left[\frac{\partial \ell^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_u'} \right] = -\sum_v \left[\frac{\mathrm{var}(\mathbf{A}_{uv})}{\mathbf{P}_{uv}^2 (1-\mathbf{P}_{uv})^2} \right] \mathbf{\Theta} \mathbf{Z}_v \mathbf{Z}_v' \mathbf{\Theta}' \tag{9}$$

$$= -\sum_v \left[\frac{1}{\mathbf{P}_{uv}(1-\mathbf{P}_{uv})} \right] \mathbf{\Theta} \mathbf{Z}_v \mathbf{Z}_v' \mathbf{\Theta}'$$

$$= -\mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} \mathbf{Z} \mathbf{\Theta} \,. \tag{10}$$

Note that $\mathrm{E}(\mathbf{A}_{uv} - \mathbf{P}_{uv})^2 = \mathrm{var}(\mathbf{A}_{uv}) = \mathbf{P}_{uv}(1-\mathbf{P}_{uv})$.

It should be noted that the gradients and hessians discussed here are inexact, as the case where $u = v$ requires an additional scaling factor. That is, $\frac{d\mathbf{P}_{uv}}{d\mathbf{Z}_u} = \mathbf{\Theta}\mathbf{Z}_v$ if $u \neq v$ and $\frac{d\mathbf{P}_{uv}}{d\mathbf{Z}_u} = 2\mathbf{\Theta}\mathbf{Z}_v$ if $u = v$. This factor is ignored for ease of notation. Also notice that all cross derivative in the hessian $\frac{\partial \ell^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_v'}$ are ignored (assumed equal to 0) because of the iid assumption.

1.2 Poisson

Now consider the multi-graph definition of $\mathcal{G}$. Here, let $\mathbf{A}_{uv} \in \mathbb{N}_0$ be the number of parallel edges from node u to node v. This type of graph is well-described by the assumption

$$\mathbf{A}_{uv} | \mathbf{Z} \sim \mathrm{Poisson}(\mathbf{Z}_u' \mathbf{\Theta} \mathbf{Z}_v) \,.$$

The log-likelihood of this model is

$$\ell(\mathbf{Z}, \mathbf{\Theta} | \mathbf{A}) = \sum_{u,v} \big[\mathbf{A}_{uv} \ln\big(\mathbf{Z}_u' \mathbf{\Theta} \mathbf{Z}_v \big) - \mathbf{Z}_u' \mathbf{\Theta} \mathbf{Z}_v - \ln(\mathbf{A}_{uv}!) \big] \tag{11}$$

$$\propto \sum_{u,v} [\mathbf{A}_{uv} \ln(\mathbf{P}_{uv}) - \mathbf{P}_{uv}] \tag{12}$$

where again $\mathbf{P}_{uv} = \mathbf{Z}_u' \mathbf{\Theta} \mathbf{Z}_v$. The gradient of this log-likelihood is

$$\frac{\partial \ell}{\partial \mathbf{Z}_u} = \sum_v \left[\frac{\mathbf{A}_{uv} - \mathbf{P}_{uv}}{\mathbf{P}_{uv}} \right] \mathbf{\Theta} \mathbf{Z}_v \tag{13}$$

$$= \mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} (\mathbf{A}_u - \mathbf{P}_u) \tag{14}$$

where

$$\mathbf{W}^{[u]} = \operatorname{diag}\left(\dots \frac{1}{\mathbf{P}_{uv}} \dots \right) . \tag{15}$$

Note that this weight matrix $\mathbf{W}^{[u]}$ is again defined over nodes $v \in \mathcal{V}$ with fixed u. The hessian of this log-likelihood is

$$\begin{aligned} \frac{\partial \ell^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_u'} &= -\sum_v \left[\frac{\mathbf{A}_{uv}}{\mathbf{P}_{uv}^2} \right] \mathbf{\Theta} \mathbf{Z}_v \mathbf{Z}_v' \mathbf{\Theta}' & (16) \\ &= -\mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} \mathbf{Y}^{[u]} \mathbf{W}^{[u]} \mathbf{Z} \mathbf{\Theta} & (17) \end{aligned}$$

where

$$\mathbf{Y}^{[u]} = \operatorname{diag}(\mathbf{A}_u) . \tag{18}$$

The expectation of the hessian is

$$\begin{aligned} \mathrm{E}\left[\frac{\partial \ell^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_u'} \right] &= -\sum_v \left[\frac{1}{\mathbf{P}_{uv}} \right] \mathbf{\Theta} \mathbf{Z}_v \mathbf{Z}_v' \mathbf{\Theta}' & (19) \\ &= -\mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} \mathbf{Z} \mathbf{\Theta} & (20) \end{aligned}$$

Note that $\mathrm{E}(\mathbf{A}_{uv}) = \mathbf{P}_{uv}$.

1.3 Normal

Now consider a graph $\mathcal{G}$ defined with real-valued edge weights. Here, let $\mathbf{A}_{uv} \in \mathbb{R}$ be the weight of an edge from node u to node v. This type of graph can be described by the assumption

$$\mathbf{A}_{uv} | \mathbf{Z} \sim \operatorname{Normal}(\mathbf{Z}_u' \mathbf{\Theta} \mathbf{Z}_v,\, \sigma^2) .$$

The log-likelihood of this model is

$$\begin{aligned} \ell(\mathbf{Z}, \mathbf{\Theta} | \mathbf{A}) &= \sum_{u,v} \left[\ln\left(\frac{1}{\sqrt{2\pi}\sigma} \right) + \frac{1}{2\sigma^2} (\mathbf{A}_{uv} - \mathbf{Z}_u' \mathbf{\Theta} \mathbf{Z}_v)^2 \right] & (21) \\ &\propto \frac{1}{2} \sum_{u,v} (\mathbf{A}_{uv} - \mathbf{P}_{uv})^2 & (22) \end{aligned}$$

where we let $\sigma^2 = 1$ and once again $\mathbf{P}_{uv} = \mathbf{Z}_u' \mathbf{\Theta} \mathbf{Z}_v$. The gradient of this log-likelihood is simply

$$\begin{aligned} \frac{\partial \ell}{\partial \mathbf{Z}_u} &= \sum_v \left[\mathbf{A}_{uv} - \mathbf{P}_{uv} \right] \mathbf{\Theta} \mathbf{Z}_v & (23) \\ &= \mathbf{\Theta}' \mathbf{Z}' (\mathbf{A}_u - \mathbf{P}_u) & (24) \\ &= \mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} (\mathbf{A}_u - \mathbf{P}_u) & (25) \end{aligned}$$

where

$$\mathbf{W}^{[u]} = \mathbf{I} \tag{26}$$

is constant for all nodes. The hessian of this log-likelihood is

$$\frac{\partial \ell^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_u'} = -\sum_v \mathbf{\Theta} \mathbf{Z}_v \mathbf{Z}_v' \mathbf{\Theta}' \tag{27}$$

$$= -\mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} \mathbf{Z} \mathbf{\Theta} \,. \tag{28}$$

The expectation of the hessian is

$$\mathrm{E}\left[\frac{\partial \ell^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_u'}\right] = -\mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} \mathbf{Z} \mathbf{\Theta} \,. \tag{29}$$

2 Estimation

2.1 Structure MLE

To estimate the parameters any of the above SBMs, we use the closed form maximum likelihood estimate (MLE) for $\mathbf{\Theta}$ and a modification of the Fisher scoring method for $\mathbf{Z}$. It is convenient that the MLE of $\mathbf{\Theta}_{ij}$ for all three models is simply the average over the adjacency matrix between nodes in community i and community j. Specifically,

$$\left[\hat{\mathbf{\Theta}}_{\text{MLE}}(\mathbf{A}, \mathbf{Z})\right]_{ij} = \frac{\mathbf{M}_{ij}}{\mathbf{n}_i \mathbf{n}_j} \tag{30}$$

where

$$\mathbf{M} = \mathbf{Z}' \mathbf{A} \mathbf{Z} \,, \qquad \mathbf{n} = \sum_u \mathbf{Z}_u \,. \tag{31}$$

2.2 Projected Fisher update

In the standard Fisher scoring method, the update rule for each row of $\mathbf{Z}$ is

$$\mathbf{Z}_u \leftarrow \mathbf{Z}_u - \mathrm{E}\left[\frac{\partial \ell^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_u'}\right]^{-1} \left[\frac{\partial \ell}{\partial \mathbf{Z}_u}\right] \tag{32}$$

$$= \mathbf{Z}_u + \left(\mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} \mathbf{Z} \mathbf{\Theta}\right)^{-1} \mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} (\mathbf{A}_u - \mathbf{P}_u) \tag{33}$$

$$= \left(\mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} \mathbf{Z} \mathbf{\Theta}\right)^{-1} \mathbf{\Theta}' \mathbf{Z}' \mathbf{W}^{[u]} \mathbf{A}_u \tag{34}$$

where $\mathbf{P}_u = \mathbf{Z} \mathbf{\Theta} \mathbf{Z}_u$ and $\mathbf{W}^{[u]}$ is given by equation 5, 15, or 26 depending on assumptions of the underlying graph.

This approach works well when the variable in question is continuous and unbounded. In our case however, $\mathbf{Z}_u$ is restricted to a feasible set of the standard basis vectors — that is, unit vectors with all but one elements equal to zero. For this reason, we propose to project the updated partition back onto the feasible set using the "hardmax" function:

$$\tilde{\mathbf{Z}}_u \leftarrow \left(\mathbf{\Theta}'\mathbf{Z}'\mathbf{W}^{[u]}\mathbf{Z}\mathbf{\Theta}\right)^{-1}\mathbf{\Theta}'\mathbf{Z}'\mathbf{W}^{[u]}\mathbf{A}_u \tag{35}$$

$$\mathbf{Z}_u \leftarrow \mathrm{hardmax}(\tilde{\mathbf{Z}}_u) \tag{36}$$

where $\mathrm{hardmax}(\mathbf{x})_i = 1$ if $i = \arg\max_j \mathbf{x}_j$ otherwise 0.

To avoid iterating over each node $u \in \mathcal{V}$, it is convenient to update the full updates all at once instead of individually for every row. Thus we define the average weight matrix

$$\overline{\mathbf{W}} = \frac{1}{|\mathcal{V}|}\sum_u \mathbf{W}^{[u]} \tag{37}$$

and apply the update rule

$$\tilde{\mathbf{Z}} \leftarrow \left[\left(\mathbf{\Theta}'\mathbf{Z}'\overline{\mathbf{W}}\mathbf{Z}\mathbf{\Theta}\right)^{-1}\mathbf{\Theta}'\mathbf{Z}'\overline{\mathbf{W}}\mathbf{A}\right]' \tag{38}$$

$$\mathbf{Z} \leftarrow \mathrm{hardmax}(\tilde{\mathbf{Z}}) \tag{39}$$

where hardmax is applied row-wise. This offers a significant increase in computational efficiency.

To ensure stability and smoothness, we also add a regularization term $\alpha\mathbf{I}$ to the hessian, where α is the regularizer strength. The MLE for $\mathbf{\Theta}$ is re-computed after each update. The full algorithm iterates over the following three steps until convergence is reached:

$$i.)\qquad \tilde{\mathbf{Z}} \leftarrow \left[\left(\mathbf{\Theta}'\mathbf{Z}'\overline{\mathbf{W}}\mathbf{Z}\mathbf{\Theta} + \alpha\mathbf{I}\right)^{-1}\mathbf{\Theta}'\mathbf{Z}'\overline{\mathbf{W}}\mathbf{A}\right]' \tag{40}$$

$$ii.)\qquad \mathbf{Z} \leftarrow \mathrm{hardmax}(\tilde{\mathbf{Z}}) \tag{41}$$

$$iii.)\qquad \mathbf{\Theta} \leftarrow \hat{\mathbf{\Theta}}_{\text{MLE}}(\mathbf{A}, \mathbf{Z}) \tag{42}$$

Let $\mathbf{Z}$ be initialized as samples from a $\mathrm{Multinomial}(1, \frac{1}{k})$ and $\mathbf{\Theta}$ be initialized as the MLE using the initial $\mathbf{Z}$. Convergence is reached when $\mathbf{Z}$ changes by less than ϵ after each update.

For this method, some tuning is necessary to identify the optimal number of communities k and ridge strength α.

2.3 Extension: Unknown number of communities

If the true number of communities is unknown, we propose to start with a large number of communities and iteratively drop those which are not "active". By "active", we mean communities containing greater than a minimum number of nodes, $n_{\min}$. Thus, the initial number of communities is set to $k_0 = \left\lfloor \frac{|\mathcal{V}|}{n_{\min}} \right\rfloor$. After each Fisher update to the partition matrix, communities with fewer than $n_{\min}$ nodes are dropped.

Dropping is done by first computing the size of each community in the updated partition. That is, the size of community i is $\mathbf{n}_i = |\{\arg\max_j \tilde{\mathbf{Z}}_{uj} = i\}|$. We then remove the i^{th} column from $\tilde{\mathbf{Z}}$ if $\mathbf{n}_i < n_{\min}$ (for every i). We refer to this procedure by the function: $\text{drop}(\tilde{\mathbf{Z}}; n_{\min})$.

To force elements of $\mathbf{n}$ to shrink toward the threshold $n_{\min}$, we augment the likelihood function with a per-node penalty:

$$\ell^*(\mathbf{Z}, \boldsymbol{\Theta}|\mathbf{A}) = \ell(\mathbf{Z}, \boldsymbol{\Theta}|\mathbf{A}) - \beta \textstyle\sum_u r(\mathbf{Z}_u) \tag{43}$$

where β is the penalty strength and r is defined as

$$r(\mathbf{Z}_u) = \mathbf{Z}_u' \boldsymbol{\Pi}^{-\gamma} \mathbf{Z}_u \tag{44}$$

where

$$\boldsymbol{\Pi} = \text{diag}(\boldsymbol{\pi}), \qquad \boldsymbol{\pi} = \frac{1}{|\mathcal{V}|} \sum_u \mathbf{Z}_u \tag{45}$$

and γ is a non-negative hyperparameter. Notice that r is larger for nodes that are assigned to small communities. Thus, a node belonging to a small community penalizes the likelihood more than a node belonging to a large community. The γ hyperparameter controls degree to which nodes are penalized. When $\gamma = 0$, the penalization is constant and has no effect; when $0 < \gamma < 1$, the penalization is dampened; when $\gamma > 1$, the penalization becomes more aggressive.

The derivatives of this penalty are

$$\frac{\partial r}{\partial \mathbf{Z}_u} \propto \boldsymbol{\Pi}^{-\gamma} \mathbf{Z}_u, \qquad \frac{\partial r^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_u'} \propto \boldsymbol{\Pi}^{-\gamma}. \tag{46}$$

and the Fisher update for $\mathbf{Z}_u$ is now

$$\begin{aligned}
\mathbf{Z}_u \leftarrow \mathbf{Z}_u - \mathrm{E}\left[\frac{\partial \ell^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_u'} - \beta \frac{\partial r^2}{\partial \mathbf{Z}_u \partial \mathbf{Z}_u'}\right]^{-1} \left[\frac{\partial \ell}{\partial \mathbf{Z}_u} - \beta \frac{\partial r}{\partial \mathbf{Z}_u}\right] && (47)\\
= \mathbf{Z}_u + \left(\boldsymbol{\Theta}'\mathbf{Z}'\mathbf{W}^{[u]}\mathbf{Z}\boldsymbol{\Theta} + \beta \boldsymbol{\Pi}^{-\gamma}\right)^{-1} \left[\boldsymbol{\Theta}'\mathbf{Z}'\mathbf{W}^{[u]}(\mathbf{A}_u - \mathbf{P}_u) - \beta \boldsymbol{\Pi}^{-\gamma} \mathbf{Z}_u\right] && (48)\\
= \left(\boldsymbol{\Theta}'\mathbf{Z}'\mathbf{W}^{[u]}\mathbf{Z}\boldsymbol{\Theta} + \beta \boldsymbol{\Pi}^{-\gamma}\right)^{-1} \boldsymbol{\Theta}'\mathbf{Z}'\mathbf{W}^{[u]}\mathbf{A}_u. && (49)
\end{aligned}$$

For convenience, we allow the penalty term $\beta \boldsymbol{\Pi}^{-\gamma}$ to replace the smoothness regularization term $\alpha \mathbf{I}$ by setting $\alpha = c\beta$ where c is a scaling constant ensuring that $c\,\text{tr}(\boldsymbol{\Pi}^{-\gamma}) = \text{tr}(\mathbf{I})$. Note that when $\gamma = 0$, the penalty is equivalent to the the smoothness regularization.

The full update procedure for this extension is

$$\begin{aligned}
&i.) \quad \tilde{\mathbf{Z}} \leftarrow \left[\left(\boldsymbol{\Theta}'\mathbf{Z}'\overline{\mathbf{W}}\mathbf{Z}\boldsymbol{\Theta} + \alpha \boldsymbol{\Pi}^{-\gamma}\right)^{-1} \boldsymbol{\Theta}'\mathbf{Z}'\overline{\mathbf{W}}\mathbf{A}\right]' && (50)\\
&ii.) \quad \mathbf{Z} \leftarrow (\text{hardmax} \circ \text{drop})(\tilde{\mathbf{Z}}; n_{\min}) && (51)\\
&iii.) \quad \boldsymbol{\Theta} \leftarrow \hat{\boldsymbol{\Theta}}_{\text{MLE}}(\mathbf{A}, \mathbf{Z}) && (52)
\end{aligned}$$

This method eliminates the need to manually identify the optimal number of communities k. Instead, just the penalty strength α must be tuned while the minimum community size $n_{\min}$ can be set heuristically.